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BSIT DA 2-1

A. Discuss briefly.

1. **What is an algorithm, and how is it related to daily tasks?**

An algorithm is a set of step-by-step instructions used to solve a problem or complete a task. In our daily life, we use algorithms when making a food dish we follow a recipe, getting ready for school, or making coffee.

2. **Why is it important for an IT student to learn algorithms?**

Is important because algorithms help IT students learn or develop how to solve problems step by step. They also help us understand how programs work and how to write better code.

B. Convert each JavaScript codes into a step-by-step algorithm using simple English or pseudocode.

1. First JavaScript code

```
for (let i = 1; i < 10; i *= 2) {
    console.log(i);
}
```

Step-by-step algorithm

1. Start at i=1.
2. Check if i is less than 10.
3. If i is less than 10, display the value i.
4. Multiply i by 2.
5. Check again if i is less than 10.
6. Repeat step 3-5, while the step 2(i < 10) or the condition is True.
7. When i is greater than 10, then the condition i < 10 is false.
8. Stop the loop.

Output is 1, 2, 4, 8

2. Second JavaScript code

```
for (let i = 0; i < 10; i++) {
    console.log(i);
}
```

Step-by-step algorithm

1. Start at i = 0.
2. Check if i is less than 10.
3. If it is less than 10, display the value of i.
4. Add 1 to i.

5. Check again if i is less than 10.
6. Repeat steps 3–5, while the step 2(i < 10) or the condition is True.
7. When i becomes 10, the condition i < 10 is false.
8. Stop the loop.

Output: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

C. Observations

1. What did you observe about the sequence of numbers produced by the first code?

The numbers increase by multiplying by 2, the sequence is 1, 2, 4, 8. It's fast because the value is doubled each time until is greater than 10, then it will stop.

2. What did you observe about the sequence of numbers produced by the second code?

The number increase by 1 each time, the sequence is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. It count numbers from 0 to 9. It starts at 0 and adds 1 each time until it reaches 10, then stop.

3. What is the main difference between the way the two loops progress?

The first one uses multiplication, the number increases much faster because it doubles the number each time, while the second using addition counts the number normally by adding 1.

4. How does the way an algorithm performs its steps affect its output?

In the first code, it doubled the number, it makes the loop shorter and faster, that's why the output is shorter. While the second code add 1 it makes the loop longer and counts the number, making the output longer.